

Positive lower density of primitive-root primes

AI-assisted research draft*

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Abstract

We give a short deduction of positive lower density in Artin's primitive root problem from a common Hecke zero-free strip and the marked prime-predecessor estimates of OpenAI. For every integer a that is neither -1 nor a square, the resulting bound is $\#\{p \leq X : \text{ord}_p(a) = p-1\} \gg_a X/\log X$. The construction uses only primes for which $p-1$ has a factor larger than $(p-1)^{1-\eta}$: all other possible index divisors then lie in the uniform complete-splitting range, while a bad large factor forces a small multiplicative order. We prove the required single-factor statistic in fixed arithmetic progressions, keeping the character multiplier separate from the invariant marked core. For any common width $0 < \delta < 1/2$ the proof gives the explicit lower coefficient $\alpha_a(Y)[-\log(1-\delta)] - \tau(Y)$, positive for all sufficiently large fixed Y . The companion strip theorem permits $\delta = 25/258$. No asymptotic formula or existence of a natural density is asserted.

1 The lower-density theorem

An integer a is *admissible* if $a \neq -1$ and a is not an integer square; in particular $|a| > 1$. For $p \nmid a$, put

$$i_p(a) = \frac{p-1}{\text{ord}_p(a)}, \quad A_a(x, 2x) = \#\{x < p \leq 2x : p \nmid a, i_p(a) = 1\},$$

and write $A_a(X)$ for the corresponding count with $p \leq X$. Primes denoted by p and q are rational primes throughout.

We use two analytic inputs. The first is the following common zero-free width, for a fixed number $0 < \delta < 1/2$:

$$L_F(s, \chi) \neq 0 \quad (\Re s > 1 - \delta), \quad (H_\delta)$$

for every cyclotomic field F containing μ_{12} and every finite-order Hecke character χ of F ; the pole of the principal character at 1 is permitted. The width is independent of F and χ . This is the form of the input in [1, Theorem 1.2]; the companion argument [3] permits $\delta = 25/258$.

The second input consists of the one-sided marked Type II and two-sided correlation estimates of [2, Sections 3 and 6], together with the graph contracts used in their proofs and the coefficient criterion in [1, Lemma 10.2]. We retain their stated bounds and invariance under removing powers of the marking primes, including the permitted divisor growth of the invariant core. Appendix A derives the only consequence needed here: a single long-prime factor statistic in each fixed reduced progression. In particular, the argument requires neither the full Poisson–Dirichlet limit nor a factorization law conditioned on nonabelian splitting.

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Let D be the signed fundamental discriminant of $\mathbb{Q}(\sqrt{a})$. Choose a fixed integer $Y_a \geq 3$ large enough that every prime factor of D is at most Y_a , and that

$$[\mathbb{Q}(\zeta_q, a^{1/q}) : \mathbb{Q}] = q(q-1) \quad (q > Y_a). \quad (1)$$

An explicit sufficient choice is given in Lemma 3.1. For every integer $Y \geq Y_a$, define

$$\alpha_a(Y) = \frac{1}{2} \prod_{3 \leq q \leq Y} \frac{q-2}{q-1} \left(1 - \mu(|D|) \prod_{\substack{q|D \\ q > 2}} \frac{1}{q-2} \right), \quad (2)$$

$$\tau(Y) = \sum_{q > Y} \frac{1}{q(q-1)}. \quad (3)$$

An empty product is one. If D is even, then $\mu(|D|) = 0$.

Theorem 1.1. *Assume (H_δ) and the marked analytic estimates stated above. For every admissible integer a and every fixed integer $Y \geq Y_a$,*

$$\liminf_{x \rightarrow \infty} \frac{A_a(x, 2x) \log x}{x} \geq \alpha_a(Y) [-\log(1-\delta)] - \tau(Y). \quad (4)$$

The same lower bound holds for $\liminf_{X \rightarrow \infty} A_a(X)/\pi(X)$. Moreover, $\alpha_a(Y) \asymp_a 1/\log Y$ and $\tau(Y) \leq 1/Y$. Consequently the right side of (4) is positive for all sufficiently large fixed Y .

Corollary 1.2. *Under the analytic inputs of [1, 2, 3], every admissible integer a is a primitive root for a set of primes of positive lower relative density. In particular, there are $c_a > 0$ and x_a such that*

$$A_a(x, 2x) \geq c_a \frac{x}{\log x} \quad (x \geq x_a).$$

One may use $\delta = 25/258$ in (4).

The proof has three steps. Fixed congruences remove all index primes at most Y . A long-prime factor statistic supplies a positive proportion of predecessors with exactly one large factor and a small cofactor. Complete splitting controls the remaining small index primes, and a product of the integers $a^j - 1$ controls the large one. The arithmetic argument is in Sections 2–5; the fixed-progression transfer of the analytic input is recorded in the appendix.

2 Admissible congruence classes

Put

$$M_Y = 8 \prod_{3 \leq q \leq Y} q$$

and let $\mathcal{B}_{a,Y}$ be the reduced classes $b \pmod{M_Y}$ satisfying

$$\chi_D(b) = -1, \quad b \not\equiv 1 \pmod{q} \quad (3 \leq q \leq Y), \quad (5)$$

where χ_D is the primitive quadratic character of discriminant D . Since $|D| \mid M_Y$, these conditions are well defined.

Lemma 2.1. *The proportion of reduced classes modulo M_Y in $\mathcal{B}_{a,Y}$ is $\alpha_a(Y)$ from (2). It is positive and satisfies $\alpha_a(Y) \asymp_a 1/\log Y$. Apart from the finitely many primes dividing aM_Y , every prime in these classes satisfies*

$$\gcd\left(i_p(a), \prod_{q \leq Y} q\right) = 1.$$

Proof. Before imposing the quadratic condition, the allowed proportion is $P_Y = \prod_{3 \leq q \leq Y} (q-2)/(q-1)$. If D is even, its 2-adic quadratic component is nontrivial on the odd classes modulo 8, and hence has mean zero. The quadratic nonresidue condition retains exactly half the allowed classes.

If D is odd, then $\chi_D(b) = \prod_{q|D} (b/q)$. At a prime $q \mid D$, the sum of (b/q) over $\mathbb{F}_q^\times \setminus \{1\}$ equals -1 . Therefore the mean of χ_D on the allowed classes is

$$\mu(|D|) \prod_{q|D} \frac{1}{q-2}.$$

Using $\mathbf{1}_{\{\chi_D=-1\}} = (1 - \chi_D)/2$ gives (2). Its bracket is strictly positive: when the Möbius factor is positive, $|D|$ has at least two distinct odd prime factors, so the product in the bracket is less than one. The exceptional value $D = 1$ would make a a square.

The identity

$$\frac{q-2}{q-1} = \left(1 - \frac{1}{q}\right) \left(1 - \frac{1}{(q-1)^2}\right)$$

and Mertens' theorem give $P_Y \asymp 1/\log Y$, since the second product converges to a positive limit. This proves the claimed size of $\alpha_a(Y)$.

For a prime in (5), Euler's criterion gives $a^{(p-1)/2} \equiv -1 \pmod{p}$. Thus $i_p(a)$ is odd. Every odd prime $q \leq Y$ is absent from $p-1$, and hence from $i_p(a)$. This also removes any obstruction caused by a being an odd perfect power. $\square$

3 Uniform complete splitting

For a prime q , write

$$E_q = \mathbb{Q}(\zeta_q), \quad K_q = E_q(a^{1/q}), \quad n_q = [K_q : \mathbb{Q}], \quad D_q = |\text{disc } K_q|.$$

Lemma 3.1. *Fix a prime $\ell \mid a$ and put $e_\ell = v_\ell(|a|)$. For every prime $q > \max(\ell, e_\ell)$,*

$$n_q = q(q-1), \quad \frac{\log D_q}{n_q} \leq 2 \log q + \log |a|. \quad (6)$$

Under (H_δ), ζ_{K_q} is zero-free in $\Re s > 1 - \delta$, apart from its pole at 1.

Proof. The prime ℓ is unramified in E_q , and the valuation of a at each prime above ℓ is e_ℓ , not divisible by q . Thus a is not a q th power in E_q . Since E_q contains μ_q , the nontrivial Kummer extension K_q/E_q has degree q . The relative discriminant divides the discriminant of the power basis for $a^{1/q}$, so

$$D_q \leq q^{q(q-2)} (q^q |a|^{q-1})^{q-1},$$

which gives (6).

To descend the zero-free strip, put $F_q = \mathbb{Q}(\mu_{12q})$ and $\widetilde{K}_q = K_q F_q$. Both $\widetilde{K}_q/F_q$ and $\widetilde{K}_q/K_q$ are abelian. The first abelian factorization writes $\zeta_{\widetilde{K}_q}$ as a product of the Hecke functions covered by (H_δ). The second writes $\zeta_{\widetilde{K}_q}/\zeta_{K_q}$ as a product of entire nonprincipal Hecke functions. A zero of ζ_{K_q} in the asserted region would therefore contradict the first factorization. This is the descent in [1, Lemma 9.1]. $\square$

Lemma 3.2. Assume (H_δ) . Fix $0 < \eta < \delta$ and $Y \geq Y_a$. Let $S_q(x)$ count primes in $(x, 2x]$ splitting completely in K_q . Then

$$\limsup_{x \rightarrow \infty} \frac{\log x}{x} \sum_{Y < q \leq (2x)^\eta} S_q(x) \leq \tau(Y). \quad (7)$$

Proof. For a nonnegative $f \in C_c^\infty((0, \infty))$, the smoothed prime-ideal formula of [1, (9.9)] gives, for any number field K of degree n and discriminant D_K having the strip,

$$\Theta_{K,f}(x) := \sum_{\mathfrak{p}} \sum_{j \geq 1} \log N\mathfrak{p} f\left(\frac{(N\mathfrak{p})^j}{x}\right) = x \int_0^\infty f(t) dt + O_f\left(x^{1-\delta}(\log |D_K| + n)\right). \quad (8)$$

The constant is independent of K . Indeed, the Mellin transform of f decays faster than any power on fixed strips, and the uniform zero count is

$$\#\{\rho : T \leq \Im \rho < T + 1\} \ll \log |D_K| + n \log(2 + |T|).$$

The zero sum is therefore bounded by the error in (8); the residue at zero and the integral on $\Re s = -1/2$ have the same bound. This records the field dependence needed when q grows. Uniformity in the original theta construction is not required: the common zero-free region and the normalized discriminant bound give uniformity here.

Given $\varepsilon > 0$, choose $f \geq 1$ on $[1, 2]$ with $\int f \leq 1 + \varepsilon$. Every completely split prime in the dyad contributes at least $n_q \log x$ to $\Theta_{K_q,f}(x)$. Equations (6) and (8) imply

$$S_q(x) \leq \frac{(1 + \varepsilon)x}{q(q-1) \log x} + O_{a,f}\left(\frac{x^{1-\delta} \log(2q)}{\log x}\right).$$

There are at most $(2x)^\eta$ terms, and $\eta < \delta$. Their total error is $O_{a,f,\eta}(x^{1-\delta+\eta}) = o(x/\log x)$. Taking the limsup and then letting $\varepsilon \downarrow 0$ proves (7). $\square$

For $p \nmid aq$, the condition $q \mid i_p(a)$ is equivalent to complete splitting in K_q : it says that $p \equiv 1 \pmod{q}$ and that a is a q th power in $\mathbb{F}_p^\times$. All counting exceptions caused by $p \mid a$ are finite.

4 A single large predecessor factor

Write $P^+(n)$ for the largest prime factor of n , with $P^+(1) = 1$. We need only the following one-dimensional consequence of the prime-predecessor estimates.

Proposition 4.1. Assume the marked analytic input in Section 1. Fix $M \geq 1$, a reduced class $b \pmod{M}$, and $1/2 < \beta < 1$. Then

$$\liminf_{x \rightarrow \infty} \frac{\log x}{x} \#\{x < p \leq 2x : p \equiv b \pmod{M}, P^+(p-1) > (p-1)^\beta\} \geq \frac{-\log \beta}{\phi(M)}. \quad (9)$$

Proof. Let $f \in C_c^\infty((\beta, 1))$ and $\Phi \in C_c^\infty((1, 2))$ satisfy $0 \leq f, \Phi \leq 1$. Put

$$T_{x,f}(n) = \sum_{q|n} f\left(\frac{\log q}{\log x}\right), \quad C_{x,f} = \sum_q \frac{1}{q} f\left(\frac{\log q}{\log x}\right).$$

The fixed-progression interior estimate proved in Proposition A.1 below, together with the prime number theorem in this fixed progression, gives

$$\begin{aligned} & \sum_{p \equiv b \pmod{M}} \Phi((p-1)/x) T_{x,f}(p-1) \\ &= \left(\frac{1}{\phi(M)} \int_1^2 \Phi(t) dt \int_\beta^1 \frac{f(u)}{u} du + o(1) \right) \frac{x}{\log x}. \end{aligned}$$

Here $C_{x,f} \rightarrow \int f(u) du/u$ by the prime reciprocal sum.

For all sufficiently large x , the left side is bounded above by the count in (9). Indeed, the compact support of f has minimum $\beta + \epsilon$ for some $\epsilon > 0$. Each counted q then satisfies $q \geq x^{\beta+\epsilon} > (p-1)^\beta$; there is at most one such factor because $\beta > 1/2$. The compact support of Φ also places p strictly inside $(x, 2x]$ for large x . Choose f and Φ increasing to one on their respective open intervals. Their integrals approach $-\log \beta$ and 1, proving the proposition. No estimate for the factorization boundary at exponent 1 is needed for this lower bound. $\square$

5 Proof of the lower-density theorem

Fix $0 < \eta < \delta$ and a permitted integer Y . By Proposition 4.1 with $\beta = 1 - \eta$, the primes in the classes $\mathcal{B}_{a,Y}$ satisfying

$$Q := P^+(p-1) > (p-1)^{1-\eta} \tag{10}$$

have lower dyadic coefficient at least

$$\alpha_a(Y)[- \log(1 - \eta)]. \tag{11}$$

For such a prime, write $p-1 = Qm$. Since $\eta < 1/2$,

$$m < (p-1)^\eta \leq (2x)^\eta, \quad Q > (p-1)^{1/2}. \tag{12}$$

Thus Q occurs only once, and every other prime divisor of $p-1$ is at most $(2x)^\eta$.

Lemma 2.1 removes every index prime $q \leq Y$. If $i_p(a) > 1$, a prime divisor of this index consequently falls into exactly one of the following two ranges.

First suppose $Y < q \leq (2x)^\eta$. Complete splitting in K_q and Lemma 3.2 show that the union of all these exceptions has upper dyadic coefficient at most $\tau(Y)$. This bound is taken over all primes, before restricting to (10) or to the chosen progressions.

The remaining possibility is $q = Q$. In that case $\text{ord}_p(a) \leq m < (2x)^\eta$. Hence p divides the nonzero integer

$$\prod_{1 \leq j \leq (2x)^\eta} |a^j - 1|.$$

Its logarithm is $O_a(x^{2\eta})$, while every counted prime is larger than x . All these exceptions together number

$$O_a\left(\frac{x^{2\eta}}{\log x}\right) = o(x/\log x), \tag{13}$$

because $\eta < 1/2$. This estimate covers the large index prime regardless of how close it is to p .

Subtracting both exceptional sets from (11) proves

$$\liminf_{x \rightarrow \infty} \frac{A_a(x, 2x) \log x}{x} \geq \alpha_a(Y)[- \log(1 - \eta)] - \tau(Y).$$

The parameters Y and η were fixed before $x \rightarrow \infty$. Now let $\eta \uparrow \delta$ to obtain (4). By Lemma 2.1, $\alpha_a(Y) \asymp_a 1/\log Y$; and the integer telescoping sum gives

$$\tau(Y) \leq \sum_{n>Y} \frac{1}{n(n-1)} = \frac{1}{Y}.$$

The coefficient is therefore positive once the fixed integer Y is sufficiently large.

Finally let c denote the right side of (4). If $c \leq 0$, the asserted global lower bound is immediate. If $c > 0$, fix J and sum the dyadic lower bounds over $(X/2^j, X/2^{j-1}]$, $1 \leq j \leq J$. The prime number theorem gives

$$\liminf_{X \rightarrow \infty} \frac{A_a(X)}{\pi(X)} \geq c \sum_{j=1}^J 2^{-j} = c(1 - 2^{-J}).$$

Letting $J \rightarrow \infty$ proves the global claim and completes Theorem 1.1. $\square$

Remark 5.1. The proof uses the common strip only to ensure $\eta < \delta$ in the sum of complete-splitting errors. Any positive common width therefore gives positive lower density. A wider strip increases the explicit coefficient, but the precise value 25/258 is not needed for positivity. The argument controls all index divisors inside a deliberately restricted family of predecessors; it does not estimate the unrestricted large-index tail required for the full Artin asymptotic.

A A fixed-progression interior statistic

This appendix proves the fixed-modulus extension actually used in Proposition 4.1. All equation and proposition numbers in this appendix attached to [2] refer to its 24 September 2026 version. Its graph and marked Type II estimates are retained as analytic inputs. We verify the changes caused by the progression, including the shared-label character cancellation.

Proposition A.1. *Fix $M \geq 1$, $(b, M) = 1$, a smooth compactly supported $f : (0, 1) \rightarrow \mathbb{R}$, and $\Phi \in C_c^\infty((0, \infty))$. With $T_{x,f}$ and $C_{x,f}$ as above, the marked estimates of [1, 2] imply*

$$\sum_{\substack{p \equiv b \\ (\bmod M)}} \Phi((p-1)/x) (T_{x,f}(p-1) - C_{x,f}) = o_{M,b,f,\Phi}(x/\log x). \quad (14)$$

The modulus is fixed before x tends to infinity.

We first establish the required twisted correlation. Use the notation of [2, Theorem 6.1]: F_x is its centered predecessor statistic, G is its invariant core, and W_1 is its marked weight. The core satisfies

$$G(v) = G(v^*), \quad |G(v)| \leq (\log x)^C d(v^*)^C$$

for fixed C , where d is the divisor function and v^* removes every prime-power factor belonging to the active marking groups.

Lemma A.2. *For a Dirichlet character $\chi \pmod{M}$, extended by zero away from units, the same estimate holds after inserting $\chi(w+1)$ at the second endpoint:*

$$\sum_{w \geq 1} \frac{\Psi(w/x)}{w} F_x(w) \chi(w+1) G(w+1) W_1(w) W_1(w+1) \ll_{A,M} (\log x)^{-A}. \quad (15)$$

The displayed invariance and divisor-growth hypotheses are imposed on G itself.

Proof. All active marking primes eventually exceed M . In the physical pairing of [2, Section 6.2], the positions obey $n' = n + D$. Multiply the target endpoint by $\chi(n')$ and the edge coefficient by $\overline{\chi(D)}$. To preserve the graph bound, expand

$$\overline{\chi(D)} = \sum_{h \bmod M} c_h e(hD/M), \quad \sum_h |c_h| \leq \sqrt{M}, \quad e(t) = e^{2\pi i t}.$$

Multiplication of an edge by $e(hD/M)$ is the unitary conjugation $U_h^{-1} A_\zeta U_h$, where $U_h v(n, \mathbf{p}) = e(hn/M) v(n, \mathbf{p})$. Each Fourier term is therefore bounded by the original [2, Proposition 6.2], applied to modulated endpoint vectors. Their suprema and norms do not increase; the first endpoint remains independent of the chosen marking lists. The residual ideal family and its comparison kernels are the original ones. Thus their hypotheses and logarithmic precision are unchanged.

Now remove the shared labels as in [2, (6.14)]:

$$n = D_R w, \quad n' = D_R(w + D_C), \quad D = D_R D_C.$$

Since the label products are units modulo M , the inserted factors reduce exactly to

$$\chi(n') \overline{\chi(D)} = \chi(D_R) \chi(w + D_C) \overline{\chi(D_R D_C)} = \chi(w + D_C) \overline{\chi(D_C)}. \quad (16)$$

No dependence on the shared product remains. Collision and square-exception estimates still hold, since all multipliers have modulus at most one.

For the raw pattern, $D_C = 1$, giving (15). For a comparison pattern, write $D_C = D_A D_B$ as in [2, (6.17)]. The factors $\overline{\chi(D_A)}$ and $\chi(D_B)$ are absorbed into the arbitrary bounded free-tuple tests in (6.18). The target sequence in (6.19) becomes $V(v) = \chi(v) G(v) W_t^b(v)$, with the same energy bound. That sequence is used only through its energy after the shared labels have been removed; the original fixed divisor-moment bounds still apply because $|\chi| \leq 1$. The bilinear Fourier and local Mellin estimates in Sections 6.3–6.6 still obtain their cancellation from the unchanged first endpoint F_x . The comparison terms are consequently negligible to the required logarithmic precision, proving the lemma. $\square$

Proof of Proposition A.1. The one-factor case of [2, Section 7] extracts the centered statistic from the preceding correlation. We give the changes for the fixed progression.

Initial mass and presieving. Impose the nonnegative condition $u + 1 \equiv b \pmod{M}$ in [2, Lemma 7.2]. A presieving condition $r \mid u + 1$ is incompatible with it if $(r, M) > 1$. Otherwise, for a long-prime product ℓ and a divisor-model term d' with $(\ell d', r) = 1$, the CRT count replacing (7.9) is

$$\sum_{\substack{u+1 \equiv b \pmod{M} \\ r \mid u+1, \ell d' \mid u}} \Phi(u/x) = \frac{x}{Mr\ell d'} \int_0^\infty \Phi(t) dt + O_{\Phi, M}(1).$$

If $(\ell d', r) > 1$, there are no solutions. The factors ℓ, d' are eventually coprime to M , since their prime factors tend to infinity. No condition on $(b - 1, M)$ is needed. The summed rounding errors and divisor-model comparisons are unchanged apart from fixed constants. The base mass gains $1/M$, while deleting sieve primes dividing M multiplies the local sieve product by $M/\phi(M)$. The net normalization is $1/\phi(M)$; the two centered main terms still cancel.

Replacing composite prime slots. Expand

$$\mathbf{1}_{\{v \equiv b \pmod{M}\}} = \frac{1}{\phi(M)} \sum_{\chi \bmod M} \overline{\chi(b)} \chi(v).$$

In each prime-to-rough replacement, retain $\chi(m)$ in the current prime slot m . The coefficient criterion in [1, Lemma 10.2], used in the proof of the one-sided estimate, is stable under this fixed twist. Indeed, cancellation against a testing character ψ is replaced by cancellation against $\chi\psi$, whose zero extension is a character of modulus at most $\text{lcm}(M, \text{mod } \psi)$. If the latter testing modulus was at most $(\log x)^B$, the new one is at most $(\log x)^{B+1}$ for large x . The coefficient bounds, support restrictions, and allowed Mellin twists are unchanged. Thus the same arbitrary logarithmic precision is available for each of the finitely many characters.

After all replacements the character product is the single factor $\chi(v)$, with $v = u + 1$. Keep it outside the invariant core in [2, (7.26)]. For fixed v it is unchanged by the allocation of active prime factors among slots. All failed-allocation, square, and candidate-deletion bounds use absolute values, so their estimates are unchanged. The final correlation (7.29) now contains $\chi(u + 1)$ and is bounded by Lemma A.2. This proves the progression version of the centered marked estimate (7.30).

Removing the marks. The prime mean-square estimate (7.33) bounds its restriction to this progression, since the summands are nonnegative. With B disjoint marking arrays, Cauchy–Schwarz therefore gives the same $O_M(B^{-1/2})$ loss as in (7.34), with the fixed number of marking groups absorbed in the constant. First let $x \rightarrow \infty$ for fixed array data, then let $B \rightarrow \infty$, and finally increase the fixed sieve depth, as in Section 7.6. The centered discrepancy tends to zero. The argument applies to smooth one-factor tests by the same partial summation and Mellin separation used in (7.1). This is (14). $\square$

References

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- [3] *A common zero-free strip for Hecke L-functions*, companion manuscript, 2026.